EFY TESTED Abhay Verma

ONE FLIP, ONE PULSE: The NE555 Trick

Turn any toggle switch into a momentary push-button with one NE555 and a handful of common parts. Within just '2 hours', you can build this 'low' complexity circuit for under ₹200!

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A toggle switch stays ON until you flip it back, but many systems expect a brief press, like a push-button. In a flight-simulator cockpit, for example, controls respond to key-strokes held for roughly 100-500 ms, and a toggle wired across the key lets the PC's auto-repeat keep firing the function, which disrupts time-sensitive controls. This NE555 circuit solves that by converting a toggle switch's change of state into a clean 1.1-second pulse, so a toggle can emulate a momentary push-button. An input from an external signal or toggle switch is turned into a sharp trigger by an AC-coupling and edge-detection stage, which triggers the NE555 one-shot to generate a single pulse of fixed duration. The output driver amplifies this pulse, and an indicator LED confirms it before it is delivered as a clean pulse output. Fig. 1 shows the author's prototype.

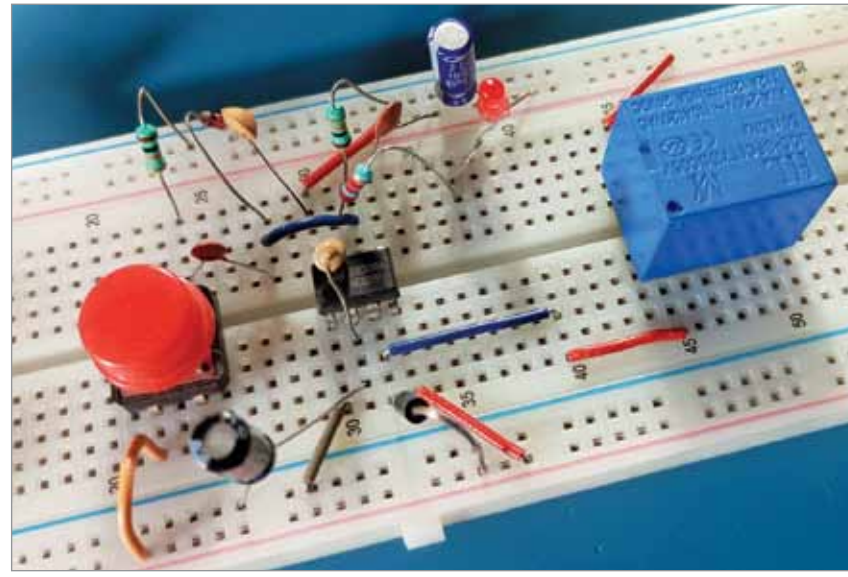


Fig. 1: Author's breadboard prototype

What will you learn?

- How the NE555 timer can turn a simple toggle input into a useful momentary control pulse.
- How to detect a rising edge using an RC network and a diode.
- How to configure the NE555 in monostable mode.
- How to calculate and adjust the pulse width using the timing resistor and capacitor.
- How to drive a relay to control an external circuit with a short pulse.
- Why a flyback diode is needed to protect the circuit from the

BILL OF MATERIALS

Reference	Component	Value/Type	Quantity	Cost (₹)
IC2	Timer IC	NE555	1	25
D1	Signal diode	1N4148	1	Miscellaneous ~175
D2	Rectifier diode	1N4007	1	
LED1	LED	5 mm, red	1	
R1, R2	Resistor	10kΩ, ¼ W, 5%	2	
R3	Resistor	2.2kΩ, ¼ W, 5%	1	
R4	Resistor	47kΩ, ¼ W, 5%	1	
C1	Electrolytic capacitor	100μF, 16 V	1	
C2	Ceramic capacitor	100nF	1	
C3	Ceramic capacitor	10nF	1	
C4	Electrolytic capacitor	22μF, 16 V	1	
C5	Ceramic capacitor	47nF	1	
RL1	Relay	5V SPDT	1	
S1	Push-to-on switch	TEST	1	
CON1-CON4	Connector	2-pin	4	
-	General-purpose PCB or breadboard, jumper wires	-	As needed	
-	Power supply	5V DC adaptor	1	Not included
Estimated total				**200**